

HOÀNG MẠNH CƯỜNG

Backend Developer (Fresher)

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SUMMARY

Backend Developer with hands-on experience developing RESTful APIs using **Node.js, NestJS and TypeScript**. Familiar with authentication, database design, bug fixing, Git workflows and Agile collaboration. Seeking a Backend Internship to strengthen backend engineering skills and contribute to real-world software products.

EXPERIENCE

HT Vietnam Joint Stock Company

Backend Developer

1/2026 – 06/2026

Hanoi, Vietnam (Onsite)

- Developed and maintained **RESTful APIs** using **Node.js, NestJS and TypeScript** for production systems.
- Designed and optimized database schemas using **PostgreSQL and MySQL** for high-performance applications.
- Implemented **JWT authentication**, role-based access control, and security best practices for backend services.
- Debugged, fixed critical backend issues, and refactored code to improve system stability and performance.
- Collaborated with cross-functional teams through code reviews, pull requests, and Agile development methodologies.

PROJECTS

VPT Code Arena | *Node.js, Socket.io, Bull/Redis Queue, Spring Boot, Next.js, PostgreSQL, Docker* | 🌐 GitHub 07/2025 – Present

- Problem:** Programming education needs to safely run untrusted user code at scale while supporting real-time multiplayer coding contests without blocking the main API.
- Solution:** Within a Spring Boot-based platform, built two dedicated **Node.js** microservices: a Socket.io service for real-time room/battle events, and a Bull-queue Judge service integrating the Judge0 API for async code execution.
- Highlight:** Decoupling real-time and code-grading workloads into separate Node.js services kept the main API responsive under concurrent socket connections and submission spikes.
- Solely designed and built the entire system (Spring Boot platform and both Node.js microservices), including a **CI/CD pipeline with GitHub Actions** (Maven build and automated tests on pull request merge).

Blockchain Copyright Management System | *NestJS, Next.js, TypeScript, MySQL, JWT, Solidity* | 🌐 GitHub 08/2024 – 02/2025

- Problem:** Digital creators lack a low-cost, tamper-proof way to register and prove ownership of their work.
- Solution:** Built **RESTful APIs** with NestJS and MySQL, and integrated Solidity smart contracts with IPFS to record and verify copyright ownership.
- Highlight:** Ownership records are immutable and independently verifiable on-chain, removing the need for a trusted middleman.

AI Call Scam Detection Backend | *Python, FastAPI, ChromaDB, Docker* | 🌐 GitHub | 📄 Paper | 🎥 Demo 07/2025 – 12/2025

- Problem:** Telecom scam calls are rising, and users need real-time risk alerts without exposing sensitive conversation data to third parties.
- Solution:** Built a FastAPI backend pipeline with vector search (ChromaDB) and local language models to classify conversations and flag scam risk in real time.
- Highlight:** Fully offline/local AI processing keeps user conversation data private while still delivering instant risk classification.

EDUCATION

East Asia University of Technology

Bachelor of Information Technology – GPA: 3.63/4.00

2022 – 08/2026

Hanoi, Vietnam

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java

Backend: Node.js, NestJS, FastAPI, RESTful API, JWT Authentication

Databases: PostgreSQL, MySQL, ChromaDB

Tools and Platforms: Git, GitHub, GitHub Actions (CI/CD), GitLab, Docker, Redis, Jira, Postman

Knowledge: Object-Oriented Programming, Database Design, Microservices, WebSocket, Agile/Scrum

ACHIEVEMENTS

- First Prize – VietFuture Competition Preliminary Round, 2025.
- Third Prize – University-level Scientific Research, 2025.
- Recipient of a 100% academic scholarship, 2025.
- Published AI-related research in the Vietnam Journal of Artificial Intelligence, 2026.